

Massimo Caruso

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Education

Montreal, Canada

Concordia University

Jan 2023 – Present

- **Major:** Software Engineering, BEng
- **Relevant Courses:** Data Structures and Algorithms, Operating Systems, Databases, Embedded Systems, Machine Learning, and Deep Learning

Experience

Lockheed Martin

Incoming May 2026 – August 2026

Database Developer (CSC Ship Integration Team), Intern

- Streamline processes and automate steps with reporting tools and data collection.
- Pull data from various sources, manipulate and consolidate the data.
- Prototype a proof-of-concept Dashboard Status Reporting Tool.
- Design and implement a user interfaces allowing users to poll the DB and perform queries.
- Document and presents tool to the team while supporting documentation for the proposal to move the proof-of-concept into a funded program.

PayFacto - Payment Technology Solutions

May 2025 – August 2025

Software Implementation, Intern

- Spearheaded data acquisition, cleaning, and preprocessing of raw merchant data sets for the deployment of MEVWeb, a province-wide cloud-based SaaS platform mandated by Revenu Québec.
- Participated in end-to-end deployment of MEVWeb software remotely and on-site, including planning, testing, and installation.
- Supported hardware transitions by uninstalling legacy MEV devices and installing MEVWeb-compatible components such as printers and routers.
- Collaborated with project delivery, field services, and sales teams to ensure seamless coordination of deployments and merchant onboarding.
- Tested MEVWeb software builds and deployment packages, reporting bugs and verifying stability before production rollout.

AtHackCTF

Nov 2024 – March 2025

Challenge Designer and Developer, Permanent Part-time

- Designed a complex RFID-based Capture the Flag (CTF) challenge using a real ATM machine and MIFARE Classic cards, which communicated with the machine's reader to simulate a security environment.
- Developed three flags requiring participants to:
 - Extract the card's PIN from memory by reverse-engineering the RFID data.
 - Manipulate the card's balance data, allowing the participant to alter the funds stored on the card.
 - Modify the card's UID to impersonate an admin and escalate privileges within the system.
- Implemented an interactive ATM interface, including buttons for navigation and a printer to issue flags upon successful completion of challenges.
- Facilitated the learning of hardware security, from memory manipulation to privilege escalation, within a real-world context.
- Prepared over 600 MIFARE Classic cards by writing custom data to each card and ensuring proper labeling and formatting for participant use.

Projects

Market Predictability (Deep Learning Project)

- Designed and implemented a PyTorch LSTM-based trading model to learn daily market leverage allocations on the Hull Tactical Market Prediction dataset, achieving an adjusted Sharpe score of **1.94**, indicating strong risk-adjusted performance with nearly twice as much return per unit of risk.
- Led systematic hyperparameter optimization across sequence length, hidden dimensions, learning rate, and regularization, identifying an optimal configuration that produced a cumulative strategy return of **122.4%** versus **116.7%** for a market benchmark.
- Developed evaluation and visualization pipelines to analyze equity curves, drawdowns, and rolling risk metrics, observing an average rolling Sharpe of approximately **1.14** with a maximum drawdown of **-25.2%**.

Embedded Wireless Access Control System

- Designed a two-node embedded access control system using ESP32 with BLE and LoRa, implementing servo-based lock actuation, tamper detection via analog sensing, and a deterministic finite state machine (armed, alarm, 2FA, disarmed) in C++ using PlatformIO.
- Implemented application-layer AES-128 encrypted alert messaging over LoRa and a BLE-based two-factor challenge-response protocol for local disarm, alongside an authenticated admin console with lockout policy and simulated biometric verification.
- Integrated multi-sensor inputs (potentiometer door position, IR motion, touch-based biometric) with encrypted wireless alerts, real-time console monitoring, and modular firmware architecture supporting offline simulation and incremental security hardening.

Food Database System

- Developed an online nutritional database integrating APIs (FatSecret, TheMealDB) to provide detailed nutritional data, recipes, and dietary metadata.
- Cleaned and validated JSON data, storing in hybrid databases (PostgreSQL + MongoDB) with optimized queries, indexing, and aggregates.
- Automated data migration between SQL and NoSQL, improving retrieval efficiency with custom recipe-name generator for FatSecret API.

Skills

Languages: (Proficient): C, Java, LaTeX, HTML/CSS, Javascript, Python, SQL; (Familiar): C++, Clojure, Erlang

Frameworks: React, Node.js, Express.js, Flask, psycopg, Arduino, TensorFlow, JUnit

Libraries: Pandas, NumPy, Matplotlib, Scikit-learn, PyTorch

Developer Tools: Git, Docker, Makefile, MongoDB, PostgreSQL, Neo4j, VS Code, Eclipse, Jupyter NB, PlatformIO

Methodologies: Agile development, Scrum, Waterfall